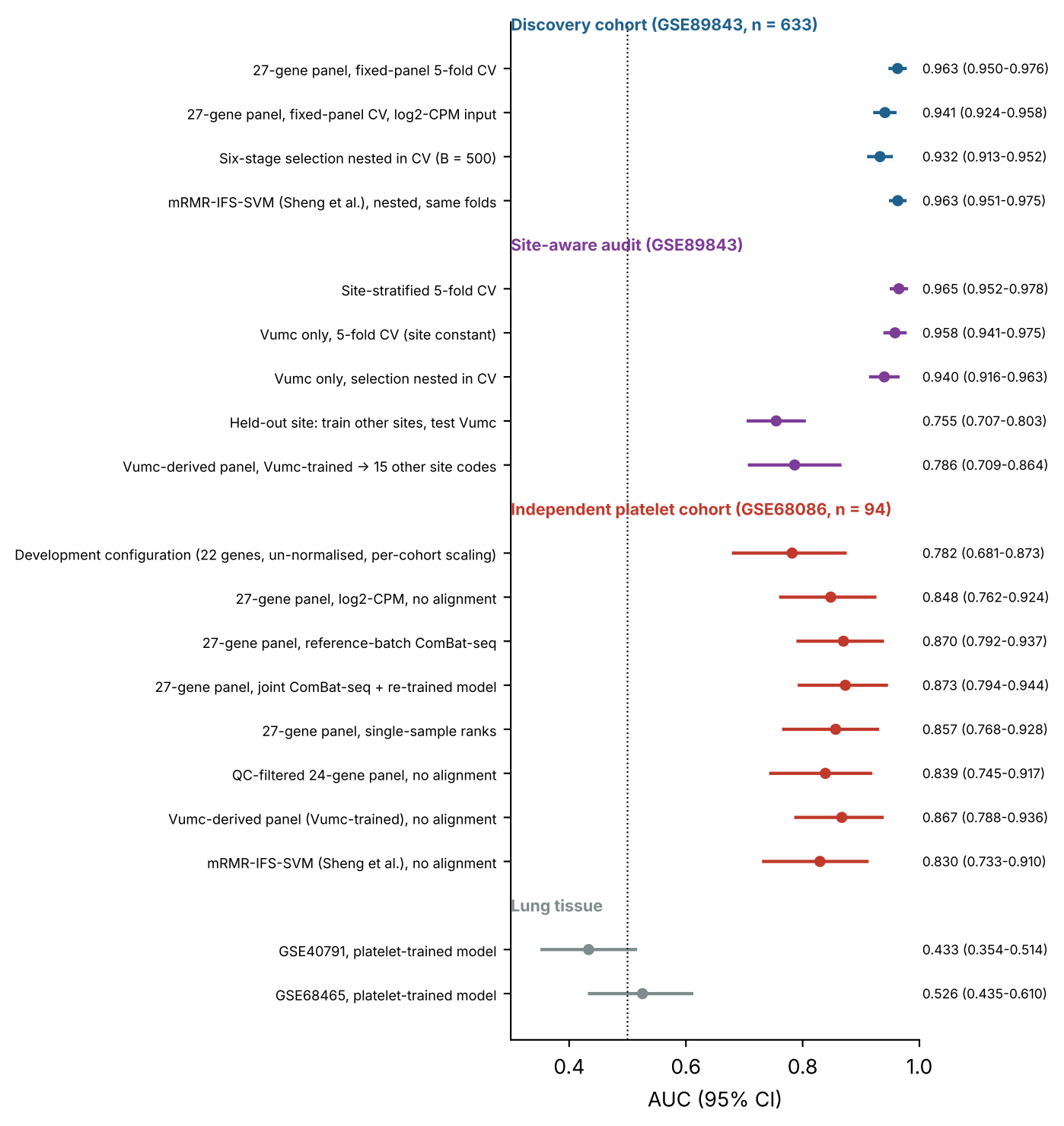
Discovery cohort (GSE89843, n = 633)
27-gene panel, fixed-panel 5-fold CV
0.963 (0.950-0.976)
27-gene panel, fixed-panel CV, log2-CPM input
0.941 (0.924-0.958)
Six-stage selection nested in CV (B = 500)
0.932 (0.913-0.952)
mRMR-IFS-SVM (Sheng et al.), nested, same folds
0.963 (0.951-0.975)
Site-aware audit (GSE89843)
Site-stratified 5-fold CV
0.965 (0.952-0.978)
Vumc only, 5-fold CV (site constant)
0.958 (0.941-0.975)
Vumc only, selection nested in CV
0.940 (0.916-0.963)
Held-out site: train other sites, test Vumc
0.755 (0.707-0.803)
Vumc-derived panel, Vumc-trained → 15 other site codes
0.786 (0.709-0.864)
Independent platelet cohort (GSE68086, n = 94)
Development configuration (22 genes, un-normalised, per-cohort scaling)
0.782 (0.681-0.873)
27-gene panel, log2-CPM, no alignment
0.848 (0.762-0.924)
27-gene panel, reference-batch ComBat-seq
0.870 (0.792-0.937)
27-gene panel, joint ComBat-seq + re-trained model
0.873 (0.794-0.944)
27-gene panel, single-sample ranks
0.857 (0.768-0.928)
QC-filtered 24-gene panel, no alignment
0.839 (0.745-0.917)
Vumc-derived panel (Vumc-trained), no alignment
0.867 (0.788-0.936)
mRMR-IFS-SVM (Sheng et al.), no alignment
0.830 (0.733-0.910)
Lung tissue
GSE40791, platelet-trained model
0.433 (0.354-0.514)
GSE68465, platelet-trained model
0.526 (0.435-0.610)
0.4
0.6
0.8
1.0
AUC (95% CI)